



AAAI-25 / IAAI-25 / EAAI-25

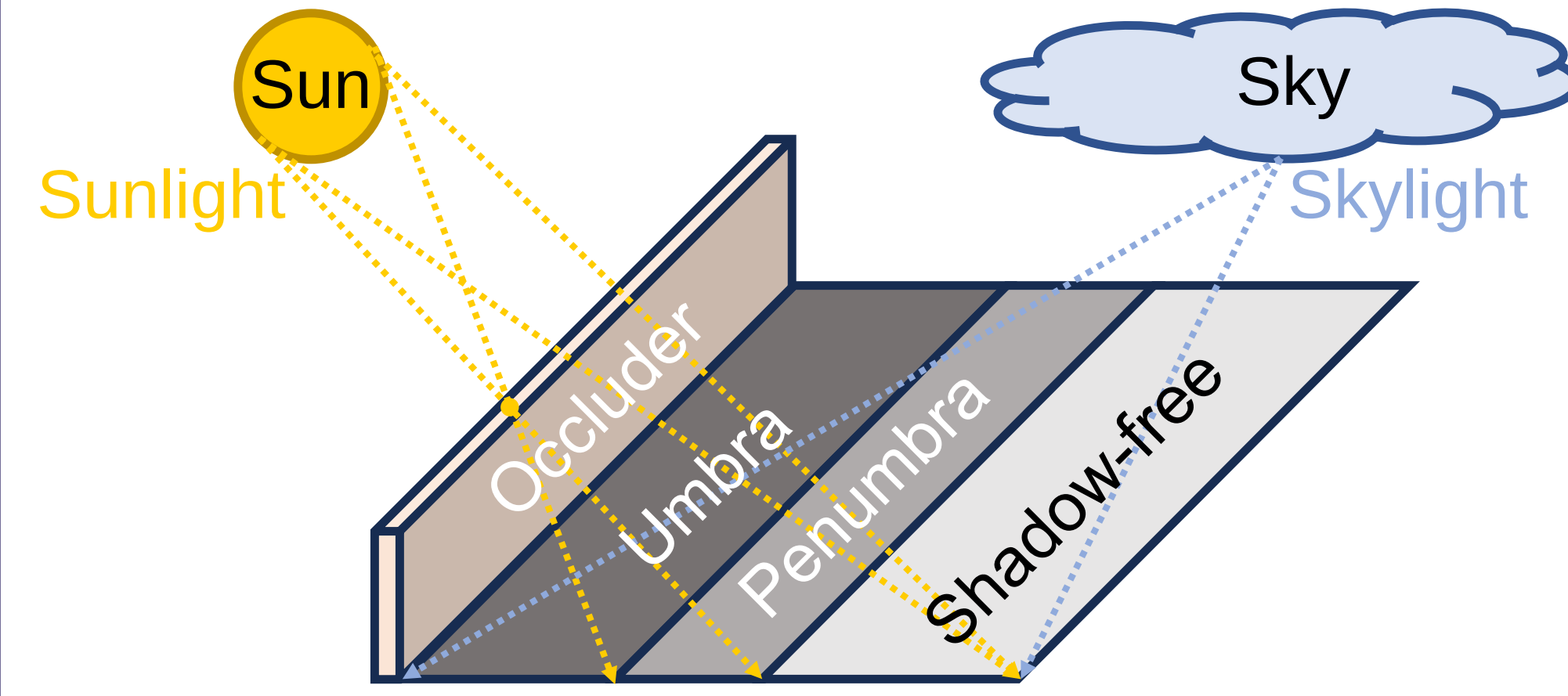
Polarization Guided Mask-Free Shadow Removal

Chu Zhou¹, Chao Xu², Boxin Shi^{2*}¹National Institute of Informatics, Japan ²Peking University, China
zhou_chu@hotmail.com, shiboxin@pku.edu.cn *Corresponding author大学共同利用機関法人 情報・システム研究機構
国立情報学研究所
National Institute of Informatics北京大学
PEKING UNIVERSITY

CONTRIBUTIONS

- **Pol-ShaRe**, the first **Polarization-guided** image **Shadow Removal** solution.
- A two-stage **pipeline** to relieve the ill-posedness of the problem.
- A neural **network** tailored to the pipeline to suppress the artifacts.

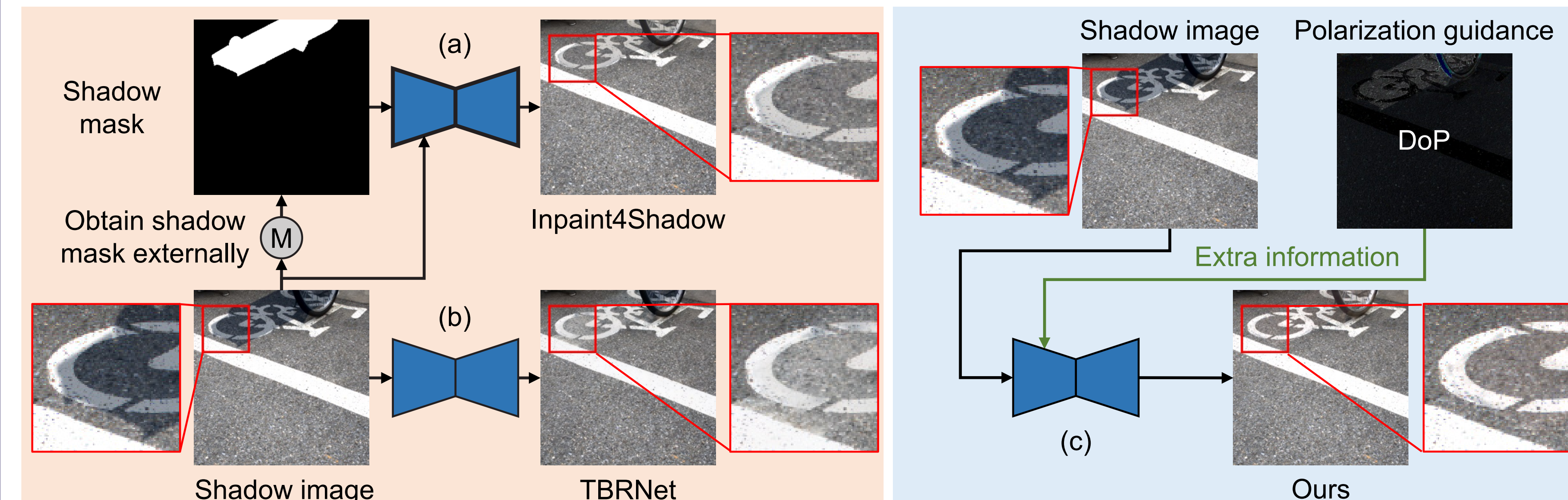
MOTIVATION



Shadow = Umbra + Penumbra

- Umbra: light is completely blocked
- Penumbra: light is partially blocked

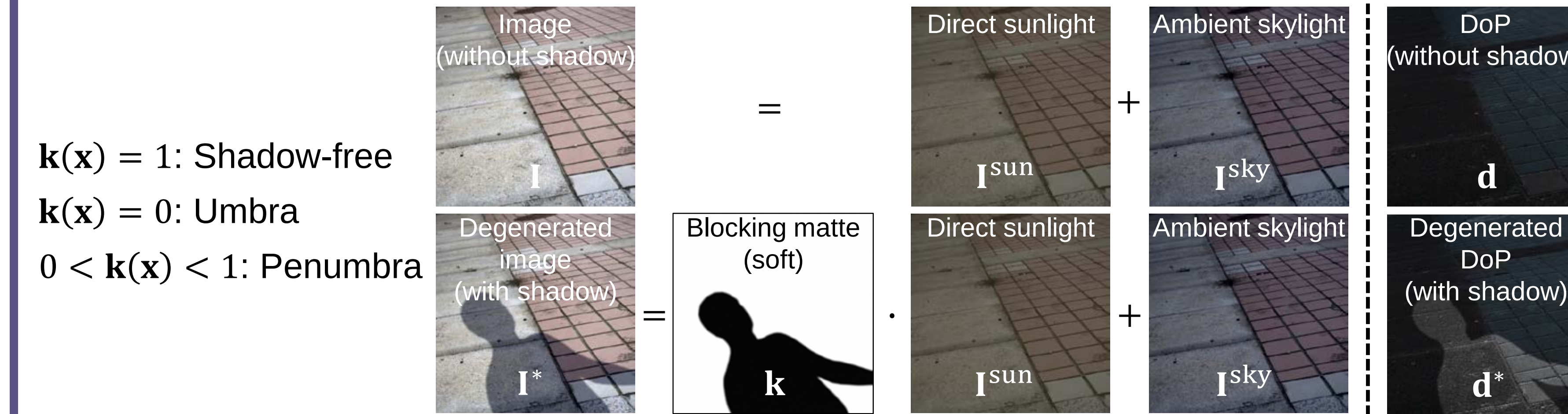
- **Shadow**: a frequently occurring phenomenon present in natural images when the **light is partially or completely blocked** in a certain region.
 - It degenerates the image quality with **color and brightness degradation**.



- Current learning-based shadow removal methods encounter **two key issues**:
 - (1) **Dependency on the shadow mask**:
 - * Many methods require an externally obtained shadow mask (e.g., Inpaint4Shadow (Li et al. 2023)), which not only introduces **additional workload**, but also leads to **degenerated performance near the shadow boundary** due to the frequently occurring inaccuracy of the mask.
 - (2) **Artifacts inside the shadow region**:
 - * There are also some methods that do not require the shadow mask (e.g., TBRNet (Liu et al. 2023b)), however, they tend to suffer from artifacts inside the shadow region, such as **false color and less-distinctive textures**.
- We find that **polarization can provide guidance** to the shadow removal process,
 - enabling us to remove shadow in a **mask-free** manner with **fewer artifacts**.

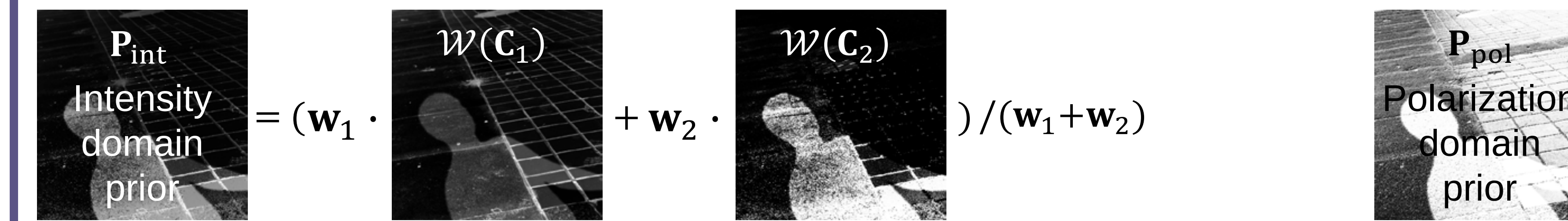
METHOD

Shadow image formation

 $k(x) = 1$: Shadow-free $k(x) = 0$: Umbra $0 < k(x) < 1$: Penumbra

- Considering the **outdoor scenes lit by daylight under sunny weather**, there are mainly two light sources: **direct sunlight** and **ambient skylight**.
- Assume for a moment some occluders block the sunlight in a certain region, which cast **shadow** in the image plane, the captured image would **degenerate**.

Dual domain priors



- **Intensity domain prior**: $P_{int} = (w_1 \cdot W(C_1) + w_2 \cdot W(C_2)) / (w_1 + w_2)$, where W is a weighting function, C_1 and C_2 are two conditions defined as below:
 - $C_1 = (1 - I_R^*) \cdot (1 - I_G^*) \cdot (1 - I_B^*)$, indicating the pixel values are relatively smaller in the shadow region compared with the shadow-free region;
 - $C_2 = (2I_B^* - I_R^* - I_G^*) / (2I_{AVG}^*)$, indicating the pixel values in the blue channel are relatively larger than the other ones in the shadow region.
- **Polarization domain prior**: $P_{pol} = W(\min(d_R^*, d_G^*, d_B^*))$, where $d^* \in [0, 1]$ is the DoP (degree of polarization) of I^* , which can be obtained using a polarizer.
 - It not only indicates the overall DoP value in the shadow region is often relatively larger compared with the shadow-free region,
 - but also in the shadow-free region there could still be a great number of pixels with relatively larger DoP values in one of the color channels.

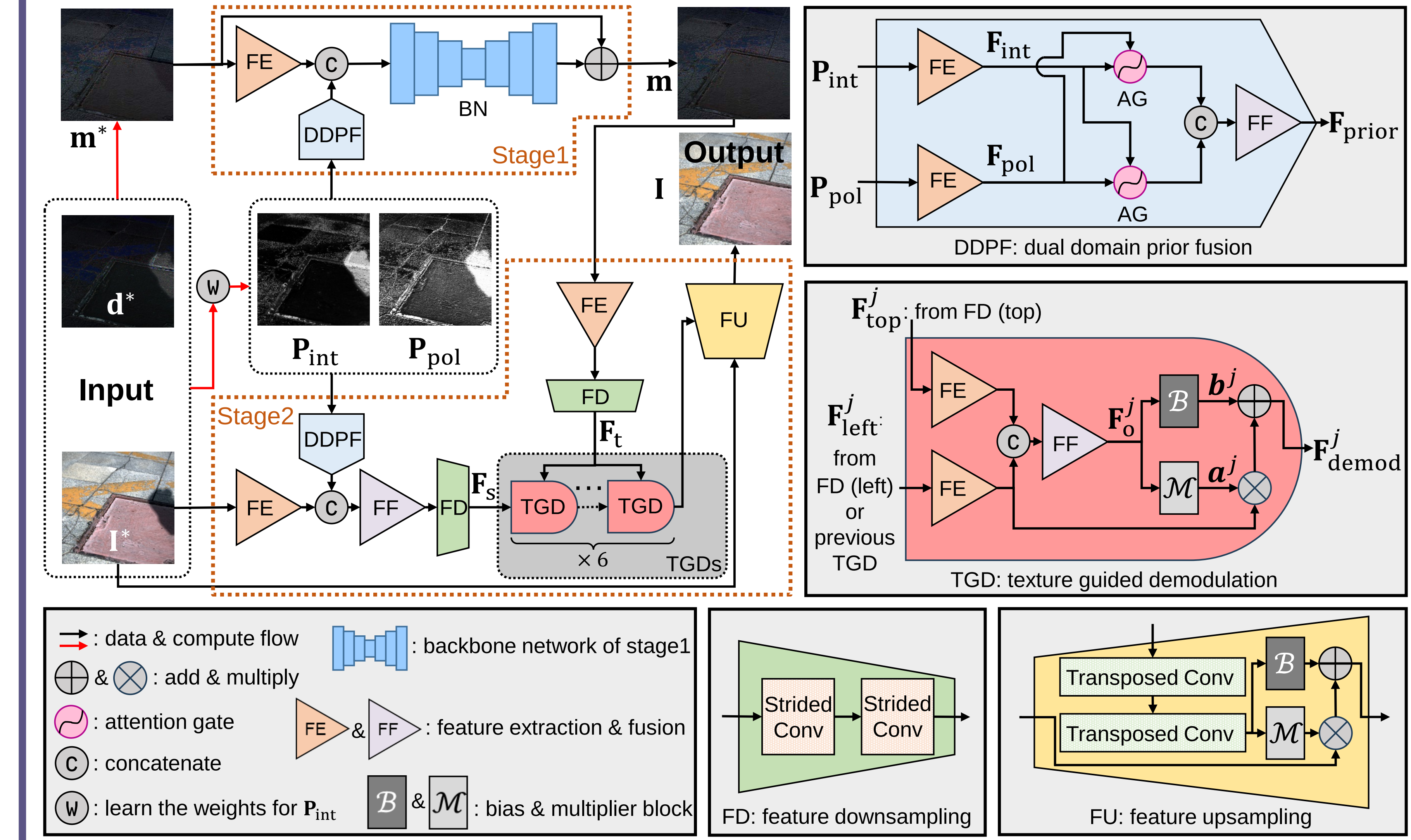
The total intensity modulated by the DoP



- **m***: More salient object contours and edges & better resistance to shadow compared with I^*
- We notice that **the total intensity modulated by the DoP has some attractive properties**, which inspire us to design a two-stage shadow removal pipeline.

METHOD

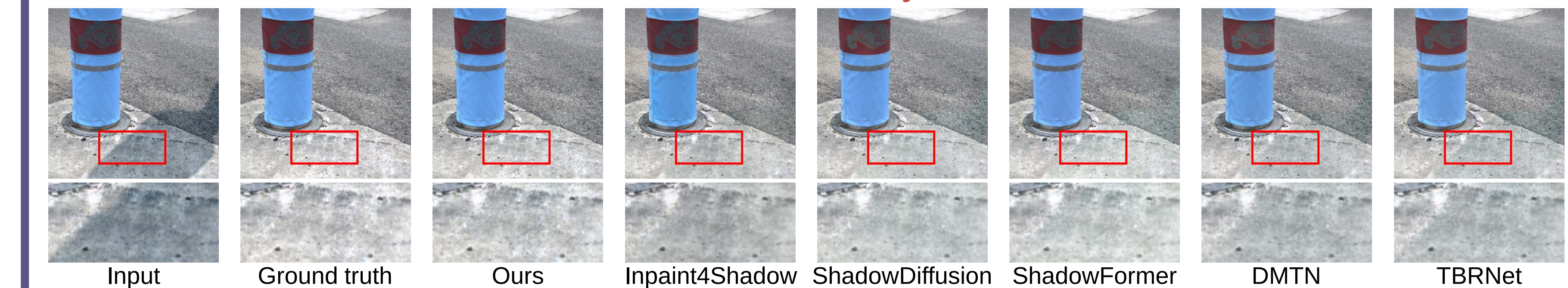
Two-stage shadow removal pipeline & Network module designs



- **Stage1**: $m = f_1(m^*, P_{int}, P_{pol})$,
 - restoring m from m^* under the guidance of P_{int} and P_{pol} , focusing on the texture and structure information.
- **Stage2**: $I = f_2(I^*, P_{int}, P_{pol}, m)$,
 - recovering I from I^* under the guidance of P_{int} and P_{pol} along with the restored m , concentrating on the brightness and color information.

EXPERIMENTS

Qualitative results on synthetic data



Qualitative results on real data

